ORIGINAL ARTICLE

Finite-sample confidence sets for nonsmooth convex-to-concave transitions

Saveliy Baturin

Independent Researcher

ARTICLE HISTORY
Compiled September 5, 2026

ABSTRACT
Many scientific curves change from increasing returns to diminishing returns, but this transition need not be smooth or uniquely located. We introduce shape-contrast inversion (SCI), a confidence procedure based on simultaneous bounds for local chord contrasts without estimating local derivatives. In fixed-design Gaussian regression, SCI covers every transition compatible with the sampled mean vector, uniformly at finite sample sizes. It allows kinks, jumps, and nonunique transitions. Extensions handle unknown constant noise, bounded heteroskedasticity, and independent replicate curves with arbitrary within-curve covariance. Across 80,000 simulations, known-scale coverage was 97.70%–98.04% at a 95% nominal level. Against a conservative pointwise-band split baseline, SCI reduced median width among nonempty outputs by 19.4%–75.7% for cusp, onset, and jump signals. Both methods remained wide for a weak logistic signal. LIDAR and replicated DNase examples illustrate the role of noise assumptions. A matrix-free implementation processes one million observations in less than one second on the benchmark machine.

KEYWORDS
Shape constraints; transition location; finite-sample inference; partial identification; multiscale contrasts

Word count: 6322 words (text, headings, and captions, including appendices; excluding references and mathematical expressions).

1. Introduction

Researchers often need to locate a change from acceleration to saturation. Examples include dose-response curves, biological growth, atmospheric profiles, disease progression, and production curves. Shape-constrained regression can estimate such a curve without choosing a detailed parametric model. In particular, an S-shaped fit is convex before a transition and concave after it [1]. The fitted transition is useful, but it does not answer a basic uncertainty question:

> Which transition locations can the data rule out while remaining valid for kinks, jumps, and flat transition regions?

This is a broadly relevant data-science problem. A reported change point is often treated as a physical threshold or an operating decision. If its uncertainty method silently

CONTACT Saveliy Baturin. Email: saveliy.xo@gmail.com

assumes a smooth and unique inflection, a narrow interval can be more misleading than no interval. The difficulty is not only computational. When curvature is zero over an interval, the data-generating curve itself does not identify one preferred transition point. A valid method must then report a set rather than invent a unique target.

Earlier work already gives important forms of inference for qualitative features. Davies et al. [2] construct finite-sample confidence regions for a regression curve and study features including inflection points. Schmidt-Hieber et al. [3] use multiscale sign statements to locate roots of operators, including curvature changes. Shape-restricted splines and bootstrap methods provide practical point and interval estimators under smoother models [4,5]. These results show that the general topic is established and important. Our goal is therefore not to claim the first confidence method for an inflection point.

We address a narrower gap. We want a direct, computable, finite-sample confidence set for the full set of convex-to-concave transition locations. The validity statement should survive nonsmooth curves, nonunique transitions, and irregular fixed designs. The output should also be usable as an uncertainty layer around a modern S-shaped point estimate.

Our method, shape-contrast inversion (SCI), starts from local chord inequalities. A positive chord residual supports convexity on its interval; a negative residual supports concavity. We construct simultaneous confidence bounds for a fixed collection of averaged residuals. Certified positive signs exclude transition locations to their left, while certified negative signs exclude locations to their right. Inverting these statements gives an outer confidence set for every compatible transition location.

The contributions are as follows.

(1) We define a design-level identified target: the closure of all transition locations admitted by at least one convex-to-concave continuation of the sampled mean vector. This separates what the fixed design identifies from interpolation.
(2) We give a direct finite-sample confidence set in fixed-design Gaussian regression. The theorem permits kinks, jumps, and nonunique transitions.
(3) We extend the construction to unknown constant variance, a declared upper bound on variance heterogeneity, and independent replicate curves with arbitrary within-curve covariance.
(4) We show localization under an explicit contrast-margin condition. The cubic special case has the familiar $(\log n/n)^{1/7}$ order; we do not claim that this order is new.
(5) We give a matrix-free implementation and compare SCI with both a published smooth-bootstrap method and a matched honest split baseline.
(6) We report weak-information cases as well as successes. A wide set is an intended answer when the data cannot support precise localization.

The rest of the paper is organized as follows. Section 2 positions the method. Sections 3–5 define SCI and prove its main guarantee. Section 6 gives the noise extensions. Sections 8 and 9 report simulations and real-data examples. Section 10 states the limits and the remaining open theory.

2. Relation to earlier work

Confidence regions and multiscale inference. The confidence-region approach of Davies et al. [2] is broad: it first constructs a set of plausible regression functions and then studies their qualitative features. Multiscale tests similarly turn many local statements into global qualitative conclusions [3,6]. SCI follows the same honest-inference principle, but it targets one feature directly. It stores only the endpoints, scale, and sufficient sums needed for local chord contrasts. This direct inversion is the source of its simple proof and its computational scale. Shape-restricted confidence bands and intervals provide closely related guarantees for monotone and convex regression [7,8]; a general account of order-restricted inference is given by Robertson et al. [9]. Multiscale change-point inference is another important example of simultaneous local testing [10].

S-shaped point estimation. The tuning-free least-squares estimator of Feng et al. [1] estimates a monotone S-shaped curve and its transition. We use its implementation as the main point estimator in our hybrid workflow. SCI does not replace that fit. It supplies a confidence set around the scientific transition target. Our coverage theorem needs the convex-to-concave restriction but not monotonicity.

Spline change-point intervals. Liao and Meyer [5] estimate a change point with shape-restricted regression splines. Their `ShapeChange` software includes a residual bootstrap interval. It is a useful comparator in its smooth spline regime. Our nonsmooth experiments deliberately include settings outside that regime, so those comparisons measure robustness rather than a failure of the published theory.

What is new here. The main object is a finite-sample outer confidence set for a possibly set-valued transition under a nonsmooth shape class. The most informative comparison is therefore not only against a point or smooth-bootstrap method, but also against another procedure with a matched finite-sample guarantee. For this purpose we implement a pointwise-band split (PBP) baseline. It is related to the confidence-region strategy of Davies et al. [2], but it is our deliberately simple baseline, not their official algorithm. Because a fixed design may only partially identify a transition, our coverage statement is for a set-valued target in the general sense of Chernozhukov et al. [11].

3. Model and target

Let

$$A \leq x_1 < \cdots < x_n \leq B, \qquad Y_i = f(x_i) + \varepsilon_i, \quad i = 1, \ldots, n,$$

where the design is fixed. In the basic model, $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$ and σ is known. Extensions relax the scale assumption.

Write $\mu = (f(x_1), \ldots, f(x_n))^\top$. The distribution of Y identifies this sampled mean vector, but it does not identify the values of f between design points. We therefore distinguish a function-level geometric object from the design-level statistical target.

Definition 3.1 (Function-level transition set)**.** For a real-valued function g on $[A, B]$, define

$$\mathcal{I}(g) = \{m \in [A, B] : g \text{ is convex on } [A, m] \text{ and concave on } [m, B]\}.$$

We call every $m \in \mathcal{I}(g)$ a compatible convex-to-concave transition of g.

The definition does not require a derivative. It permits an endpoint transition, a kink, and a one-sided jump at the transition. It also permits a flat-curvature interval, in which case more than one m is compatible.

Definition 3.2 (Design-level identified transition set)**.** Let

$$\mathcal{G}_x(\mu) = \{g : [A, B] \to \mathbb{R} : g(x_i) = \mu_i,\ i = 1, \dots, n\}.$$

The target identified by the fixed-design mean vector is

$$\mathcal{I}_x(\mu) = \mathrm{cl}\{m \in [A, B] : m \in \mathcal{I}(g) \text{ for some } g \in \mathcal{G}_x(\mu)\}\,. \tag{1}$$

The closure in (1) is deliberate. If endpoint jumps are allowed, a function-level set $\mathcal{I}(g)$ can be half-open. A closed target is the natural object for the closed confidence set reported below.

Lemma 3.3 (Geometry of the target)**.** *If $m_1 < m_2$ both belong to $\mathcal{I}(g)$, then g is affine on $[m_1, m_2]$ and $[m_1, m_2] \subseteq \mathcal{I}(g)$. Therefore, whenever $\mathcal{I}(g)$ is nonempty, it is an interval, possibly a singleton, but it need not be closed.*

The target can be computed directly from adjacent divided slopes. This is useful for checking simulations against the same set-valued object as the coverage theorem.

Proposition 3.4 (Linear-time target calculation)**.** *Let $s_i = (\mu_{i+1} - \mu_i)/(x_{i+1} - x_i)$. A point $m \in (x_k, x_{k+1})$ is admitted by a convex-to-concave continuation if and only if the slopes through $x_1, \dots, x_k$ are nondecreasing, the slopes through $x_{k+1}, \dots, x_n$ are nonincreasing, and there is a value v such that*

$$\mu_k + s_{k-1}(m - x_k) \le v \le \mu_{k+1} - s_{k+1}(x_{k+1} - m), \tag{2}$$

where a missing boundary inequality is omitted. At $m = x_i$, the analogous prefix and suffix slope conditions include the fixed value μ_i. When both neighboring slopes exist, eliminating v gives the directly checkable condition

$$(s_{k-1} - s_{k+1})t \le s_k - s_{k+1}, \qquad t = \frac{m - x_k}{x_{k+1} - x_k}.$$

Consequently, all feasible gap intervals and the endpoints of $\mathcal{I}_x(\mu)$ can be found in $O(n)$ operations. Moreover, $\mathcal{I}_x(\mu)$ is either empty or one closed interval.

Lemma 3.3 separates statistical uncertainty from identification. Even with no noise, a curve that is affine over $[m_1, m_2]$ does not select one transition inside that interval. A finite design adds interpolation uncertainty: several continuous or discontinuous shape-constrained extensions may agree at all observed locations. Our confidence statement therefore covers the whole set $\mathcal{I}_x(\mu)$, which depends only on the identified mean vector.

3.1. Chord contrasts

For design indices $L < M < R$, define the chord residual

$$Q_{L,M,R}(f) = \frac{x_R - x_M}{x_R - x_L} f(x_L) + \frac{x_M - x_L}{x_R - x_L} f(x_R) - f(x_M). \tag{3}$$

Convexity makes (3) nonnegative; concavity makes it nonpositive. Single triples are noisy, so we average nonnegative collections of chord residuals. Let $\mathcal{T}$ be a fixed finite family. Each $T \in \mathcal{T}$ defines a row vector $w_T \in \mathbb{R}^n$, a support interval $[a_T, b_T]$, and a contrast mean

$$Z_T = w_T^\top Y, \qquad \mu_T = w_T^\top \mu.$$

The weights are chosen so that $\mu_T \geq 0$ when a compatible extension g is convex on the support, and $\mu_T \leq 0$ when it is concave there. The family is fixed before seeing Y. Appendix A gives the construction used in the experiments.

4. Shape-contrast inversion

Let $M = |\mathcal{T}|$ and let Φ be the standard normal distribution function. For a desired error probability α, set

$$c_{\alpha,M} = \Phi^{-1}\left(1 - \frac{\alpha}{2M}\right). \tag{4}$$

For every contrast, form the simultaneous interval

$$[\ell_T, u_T] = \left[Z_T - c_{\alpha,M}\sigma \left\|w_T\right\|_2, Z_T + c_{\alpha,M}\sigma \left\|w_T\right\|_2\right]. \tag{5}$$

A contrast is certified positive when $\ell_T > 0$ and certified negative when $u_T < 0$. Define

$$\widehat{L} = \max\bigl(\{A\} \cup \{a_T : \ell_T > 0\}\bigr), \tag{6}$$

$$\widehat{U} = \min\bigl(\{B\} \cup \{b_T : u_T < 0\}\bigr). \tag{7}$$

The SCI output is

$$\widehat{\mathcal{C}}(Y) = \begin{cases} [\widehat{L}, \widehat{U}], & \widehat{L} < \widehat{U}, \\ \varnothing, & \widehat{L} \geq \widehat{U}. \end{cases} \tag{8}$$

The construction has a simple interpretation. A reliably convex interval cannot lie wholly after a valid transition, so its left endpoint gives a lower limit. A reliably concave interval cannot lie wholly before a valid transition, so its right endpoint gives an upper limit. Only simultaneous sign statements are inverted.

Figure 1 gives a deliberately simple view of the construction. It suppresses the calibration details and keeps the three ideas needed on first reading: chord signs, one-sided exclusions, and intersection. The figure explains the logic of the algorithm. The coverage claim comes from Theorem 5.1.

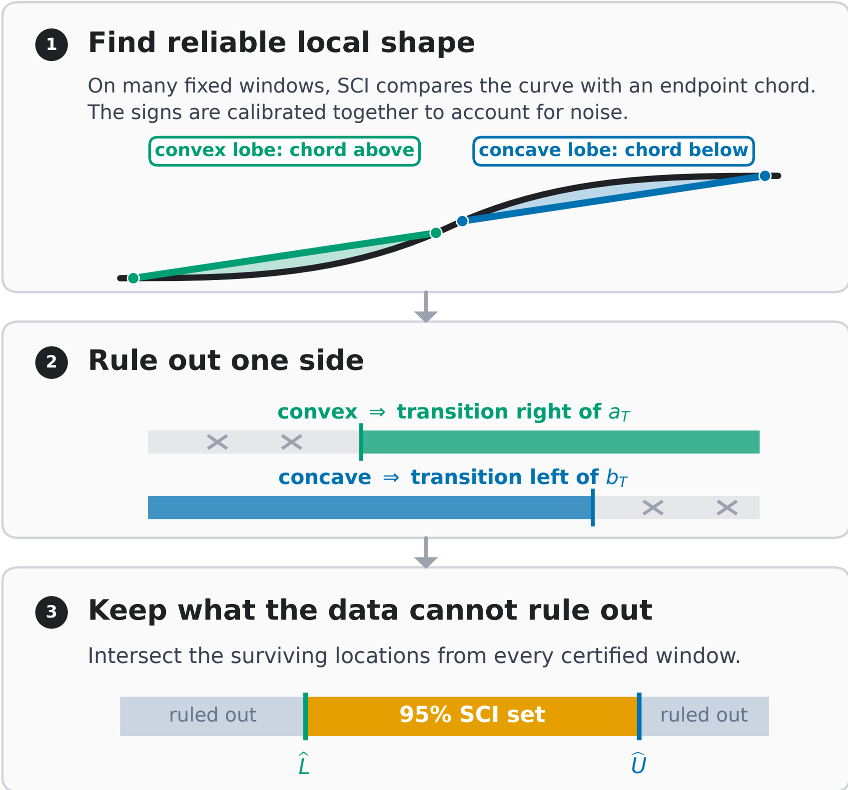

Figure 1. A plain-language overview of SCI. (1) Endpoint chords create signed local residual lobes: the chord is above a convex segment and below a concave segment. The implementation averages these residuals and calibrates all fixed windows together. (2) Certified convexity rules out transition locations on the left, while certified concavity rules out locations on the right. (3) Intersecting all surviving locations gives the SCI confidence set. This schematic illustrates the inversion logic; it is not a substitute for the coverage proof.

5. Finite-sample theory

Theorem 5.1 (Known-scale finite-sample coverage). *Suppose $\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$, the finite contrast family $\mathcal{T}$ is fixed independently of Y, and every row has the sign property stated above. Then, uniformly over every mean vector μ with $\mathcal{I}_x(\mu) \neq \varnothing$,*

$$\underset{\mu,\sigma}{\mathrm{P}}\{\mathcal{I}_x(\mu) \subseteq \widehat{\mathcal{C}}(Y)\} \geq 1 - \alpha.$$

Consequently, $\mathrm{P}_{\mu,\sigma}\{\widehat{\mathcal{C}}(Y) = \varnothing\} \leq \alpha$ whenever the target is nonempty.

The guarantee is finite-sample and nonasymptotic for the stated Gaussian model. It covers all design-compatible locations at once and therefore remains meaningful when the target is set-valued.

Proposition 5.2 (Minimal interval from the certified signs). *Condition on the collection of certified positive and negative signs. If $\widehat{L} < \widehat{U}$, then $[\widehat{L}, \widehat{U}]$ is the closure of all candidate locations not excluded by those sign implications. If $\widehat{L} \geq \widehat{U}$, no candidate location survives.*

Proposition 5.2 is a logical minimality result. It does not say that SCI is the shortest possible confidence procedure. A different joint band or additional model assumptions can give a smaller set.

5.1. Localization under a contrast margin

Coverage alone does not ensure that the set is informative. The next result states the signal strength needed for localization. It also makes clear why weak smooth transitions can produce wide sets.

Assumption 5.3 (Contrast margin at scale d)**.** Suppose $\mathcal{I}_x(\mu)$ is nonempty and write $m_- = \inf \mathcal{I}_x(\mu)$ and $m_+ = \sup \mathcal{I}_x(\mu)$. For a scale $d > 0$, suppose the family contains

(i) a left contrast T_L with $a_{T_L} \ge m_- - Kd$, $\mu_{T_L} \ge Bd^\gamma$, and $\|w_{T_L}\|_2 \le C_w/\sqrt{nd}$; and
(ii) a right contrast T_R with $b_{T_R} \le m_+ + Kd$, $\mu_{T_R} \le -Bd^\gamma$, and $\|w_{T_R}\|_2 \le C_w/\sqrt{nd}$.

Here $B, C_w, K > 0$ and $\gamma > 0$ do not depend on n.

Theorem 5.4 (Margin-based localization)**.** *Under Theorem 5.1 and Assumption 5.3, let $0 < \eta < 1 - \alpha$ and suppose*

$$Bd^\gamma \ge \frac{\sigma C_w}{\sqrt{nd}} \left(c_{\alpha,M} + z_{1-\eta/2}\right), \tag{9}$$

where $z_q = \Phi^{-1}(q)$. Then, with probability at least $1 - \eta$,

$$\widehat{\mathcal{C}}(Y) \subseteq [m_- - Kd, m_+ + Kd].$$

Moreover, with probability at least $1 - \alpha - \eta$,

$$\mathcal{I}_x(\mu) \subseteq \widehat{\mathcal{C}}(Y) \subseteq [m_- - Kd, m_+ + Kd].$$

In particular, the length bound $|\widehat{\mathcal{C}}(Y)| \le \operatorname{diam}(\mathcal{I}_x(\mu)) + 2Kd$ holds with probability at least $1 - \eta$.

Balancing the two sides of (9) gives the detection scale

$$d_n \asymp \left\{ \frac{\sigma^2 (c_{\alpha,M} + z_{1-\eta/2})^2}{B^2 n} \right\}^{1/(2\gamma+1)}. \tag{10}$$

When M grows polynomially with n, $c_{\alpha,M}^2 = O(\log n)$. Because η can be chosen for any requested tail probability, with the multiplicative constant allowed to depend on that probability, the positive excess length is conditionally

$$\{|\widehat{\mathcal{C}}(Y)| - \operatorname{diam}(\mathcal{I}_x(\mu))\}_+ = O_{\mathrm{P}}\left\{(\sigma^2 \log n/(B^2 n))^{1/(2\gamma+1)}\right\}.$$

This conclusion is conditional on the explicit margin and contrast availability at a scale within a fixed factor of (10). It is not yet a broad minimax theorem for every S-shaped function class.

5.2. Cubic transition

The standard smooth reference case is a cubic crossing. Let $x_i = i/(n+1)$ and

$$f(x) = c_0 + c_1(x - m_0) - B(x - m_0)^3, \qquad B > 0.$$

For the dyadic contrast family in Appendix A, define

$$h_* = \left\{ \frac{\sigma(c_{\alpha,M_n} + z_{1-\eta/2})}{\sqrt{6}B\sqrt{n+1}} \right\}^{2/7}. \tag{11}$$

Corollary 5.5 (Cubic localization order)**.** *Let $0 < \eta < 1 - \alpha$. Put $q_* = (n+1)h_*$, assume $q_* \geq 1$, and let q be the smallest dyadic integer with $q \geq q_*$. If $3q \leq n$ and $m_0 \in [8h_*, 1 - 8h_*]$, then*

$$\underset{f}{\mathrm{P}}\{|\widehat{\mathcal{C}}(Y)| < 16h_*\} \geq 1 - \eta,$$

and

$$\underset{f}{\mathrm{P}}\{\mathcal{I}_x(\mu) \subseteq \widehat{\mathcal{C}}(Y),\ |\widehat{\mathcal{C}}(Y)| < 16h_*\} \geq 1 - \alpha - \eta.$$

Thus

$$|\widehat{\mathcal{C}}(Y)| = O_{\mathrm{P}}\left\{ \left(\frac{\sigma^2 \log n}{B^2 n} \right)^{1/7} \right\}.$$

For the three frozen separation multipliers, $M_n \leq 6n + 6(1 + \log_2 n)$. Hence the polynomial-size condition used for the displayed stochastic order holds directly.

The $n^{-1/7}$ power is the known regular order for localizing a root of a second derivative. Related orders appear in Schmidt-Hieber et al. [3] and Feng et al. [1]. The value of Corollary 5.5 is compatibility: SCI keeps finite-sample validity over a wider nonsmooth class without losing this reference order in the cubic case.

6. Unknown and nonconstant noise

The inversion only needs simultaneous bounds for the contrast means. This modularity gives three useful extensions.

6.1. Unknown constant scale

Let $X \in \mathbb{R}^{n\times r}$ be a full-rank nuisance design fixed before Y is observed, where $0 \leq r < n$, and let $R_X = I - P_X$ be its residual projector. Write $q_{\nu,\eta}$ for the lower η quantile of a chi-squared random variable with ν degrees of freedom, and define

$$\overline{\sigma} = \left\{ \frac{\|R_X Y\|_2^2}{q_{n-r,\eta}} \right\}^{1/2}. \tag{12}$$

Theorem 6.1 (Honest upper scale bound)**.** *If $Y = \mu + \varepsilon$ with arbitrary deterministic $\mu \in \mathbb{R}^n$, $\varepsilon \sim N(0, \sigma^2 I)$, and $0 < \eta < \alpha < 1$, then*

$$\inf_{\mu, \sigma > 0} \mathrm{P}_{\mu,\sigma} (\overline{\sigma} \geq \sigma) \geq 1 - \eta.$$

If (5) uses $\overline{\sigma}$ and error budget $\alpha - \eta$, the resulting SCI *set covers $\mathcal{I}_x(\mu)$ with probability at least $1 - \alpha$ whenever that target is nonempty.*

No independence between the scale estimate and the contrasts is required. The price is conservatism: the residual norm can contain unremoved signal. Our study implementation uses fixed consecutive blocks of size two. For a monotone sampled mean with total variation at most V, its paired-block residual obeys

$$\|R_X \mu\|_2^2 = \frac{1}{2} \sum_j (\mu_{2j} - \mu_{2j-1})^2 \leq \frac{V^2}{2}.$$

Since the residual degrees of freedom grow linearly with n, the corresponding contribution to $\overline{\sigma}^2$ is $O(n^{-1})$.

6.2. Bounded heteroskedasticity

Now let the errors be independent with $\varepsilon_i \sim N(0, \sigma_i^2)$. A finite-sample result requires some information about the unobserved variance pattern. We make that information explicit.

Assumption 6.2 (Variance-ratio bound)**.** For a declared $\kappa \geq 1$,

$$\max_i \sigma_i^2 \leq \kappa \bar{\sigma}^2, \qquad \bar{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \sigma_i^2.$$

Use a fixed partition into blocks of size two or three, and let R remove the constant within each block. Set $Q = \|RY\|_2^2$ and

$$d_{n,\kappa,\eta} = 1 - 2\sqrt{\frac{2\kappa \log(1/\eta)}{n}}, \tag{13}$$

$$\widehat{\sigma}^2_{\max} = \frac{2\kappa Q}{n d_{n,\kappa,\eta}}. \tag{14}$$

Theorem 6.3 (Heteroskedastic scale envelope)**.** *Under Assumption 6.2, if $0 < \eta < \alpha < 1$ and $d_{n,\kappa,\eta} > 0$, then*

$$\mathrm{P}\left(\widehat{\sigma}^2_{\max} \geq \max_i \sigma_i^2\right) \geq 1 - \eta$$

uniformly over the mean vector and all variance vectors satisfying the assumption. Replacing each contrast standard error by $\widehat{\sigma}_{\max} \|w_T\|_2$ and using error budget $\alpha - \eta$ therefore gives at least $1 - \alpha$ coverage of $\mathcal{I}_x(\mu)$ whenever that target is nonempty.

The proof uses Anderson's inequality for a translated Gaussian norm [12] and a weighted chi-squared lower-tail inequality [13]. The method is a sensitivity analysis: its claim is only as credible as the chosen κ. Larger values correctly give wider sets.

6.3. Independent replicate curves

Suppose we observe $N_{\text{rep}} \geq 2$ independent curves

$$Y_r \sim N(\mu, \Sigma), \qquad r = 1, \ldots, N_{\text{rep}},$$

where the common covariance Σ may be non-diagonal or singular. For a contrast row w_j, let $Z_{rj} = w_j^\top Y_r$, with sample mean $\bar{Z}_j$ and sample standard deviation s_j across replicates.

Theorem 6.4 (Replicate Student band)**.** *For M fixed contrasts, the intervals*

$$\bar{Z}_j \;\pm\; t_{N_{\text{rep}}-1,1-\alpha/(2M)} \frac{s_j}{\sqrt{N_{\text{rep}}}}, \qquad j = 1, \ldots, M,$$

cover all contrast means simultaneously with probability at least $1-\alpha$. Their sign inversion therefore satisfies $\mathrm{P}\{\mathcal{I}_x(\mu) \subseteq \widehat{\mathcal{C}}\} \geq 1 - \alpha$ whenever the target is nonempty.

Dependence and unequal variance across design locations are allowed inside one replicate. Independence is required across replicates, which is the correct sampling unit in repeated-curve studies [14].

7. Computation and matched honest baseline

7.1. Matrix-free implementation

Writing all contrast rows as an $M \times n$ matrix is unnecessary. On a uniform design, block averages and chord residuals can be evaluated from prefix sums. The implementation stores only scale metadata and start indices. The full contrast family used in the experiments has separation ratios $(1, 2, 4)$ across dyadic scales. The public function retains (1) as its minimal default.

Table 1 gives a deterministic scaling benchmark. Times are hardware-specific, while storage and dense counterfactual sizes follow directly from the constructed family. At $n = 10^6$, the implementation evaluates 5,999,750 contrasts in 0.73 seconds and stores 53.4 MiB of contrast-family metadata. This number excludes the output array, prefix sums, temporary workspace, and Python overhead. A dense double matrix would require about 44,702 GiB.

Table 1. Matrix-free scaling for the full $(1, 2, 4)$ separation family. "Metadata" counts the compact numerical arrays that define the family, not total peak process memory.

n	Contrasts	Build (s)	Evaluate (s)	Metadata (MiB)	Dense (GiB)
1,000	5,872	0.0006	0.0015	0.05	0.04
10,000	59,831	0.0009	0.0039	0.53	4.46
100,000	599,790	0.0042	0.0287	5.34	446.88
1,000,000	5,999,750	0.0346	0.7318	53.40	44,701.62

7.2. Pointwise-band split baseline

To compare two methods with the same type of guarantee, we construct a generic pointwise confidence box

$$\mathcal{B}(Y) = \prod_{i=1}^{n} [Y_i - z_{1-\alpha/(2n)}\sigma, Y_i + z_{1-\alpha/(2n)}\sigma].$$

The PBP algorithm finds the longest prefix for which the box contains a discretely convex sequence and, in a separate feasibility problem, the earliest suffix for which it contains a discretely concave sequence. The two witness sequences need not meet at the same transition value. Thus PBP is a conservative discrete split relaxation, not an exact projection onto the same continuous function class as SCI. The declared domain $[A, B]$ is supplied to the implementation: a fully feasible concave suffix extends the returned set to A, and a fully feasible convex prefix extends it to B.

Proposition 7.1 (Finite-sample coverage of PBP)**.** *Under the known-scale Gaussian model on* $[A, B]$*, let* $\widehat{\mathcal{C}}_{\mathrm{PBP}}(Y)$ *be the closed interval returned by* PBP *with domain* $[A, B]$*. Then, for every sampled mean vector,*

$$\mathop{\mathrm{P}}_{\mu,\sigma}\{\mathcal{I}_x(\mu) \subseteq \widehat{\mathcal{C}}_{\mathrm{PBP}}(Y)\} \geq 1 - \alpha.$$

The comparison measures the gain over this simple conservative baseline; it is not a claim against the best possible pointwise-band projection.

8. Simulation experiments

8.1. Design

We use four benchmark signals from the S-shaped estimation study of Feng et al. [1]: a cusp, a one-sided onset, a jump, and a smooth logistic curve. The first three stress nonsmooth behavior; the logistic signal is the smooth reference case. Designs are either uniform or sampled from a Beta$(4, 8)$ distribution. Sample sizes are $n = 500$ and $n = 1000$, the Gaussian standard deviation is 0.1, and the nominal confidence level is 95%.

The study was staged. An initial comparison used 200 responses per cell and included the S-shaped least-squares point estimator (`Sshaped` 1.2) and the residual-bootstrap interval from `ShapeChange` 1.5 with 1,000 bootstrap samples. We then froze the SCI configuration and ran a fresh high-precision audit with 5,000 responses in each of 16 cells, or 80,000 responses in total. Finally, a separate 200-response-per-cell experiment compared SCI with the matched honest PBP baseline. Failures and empty sets were never discarded.

8.2. High-precision coverage audit

Table 2 summarizes the 80,000-response audit. A deterministic post-audit calculation of $\mathcal{I}_x(\mu)$ changed two of the 80,000 classifications that had originally checked only the generating point $m_0 = 0.3$; no responses were regenerated. Coverage of the full identified set is stable between 0.9770 and 0.9804. The simultaneous contrast band itself covers all contrast means in 0.9524–0.9618 of responses, close to the intended 0.95 familywise level. The larger transition-set coverage is expected because not every missed contrast bound changes the inverted set.

Table 2. High-precision known-scale audit. Each row summarizes four fixed design–sample-size cells with 5,000 responses per cell.

Signal	Coverage range	Median-width range	Largest empty rate
Cusp	0.9772–0.9804	0.0712–0.1737	0.0182
Onset	0.9770–0.9786	0.6703–0.8149	0.0066
Jump	0.9770–0.9798	0.0020–0.0100	0.0228
Logistic	0.9776–0.9798	0.9990–1.0000	0.0006

The widths contain the main scientific message. The jump is localized very sharply, the cusp moderately, and the one-sided onset only partly. The weak logistic curvature does not produce reliable local signs, so the answer is almost the whole design range. This is not a computational failure. It is the honest statement that the chosen data and noise level do not identify the transition precisely.

One pre-specified diagnostic, frozen before the high-precision audit, required zero empty sets. It failed: empty-set frequencies reached 2.28%. Theorem 5.1 only requires this probability to be at most 5% for a nonempty target, and the Wilson upper bounds remain below 5% in all cells. We nevertheless report the frozen diagnostic as failed rather than redefining it after seeing the results.

8.3. Published comparator and hybrid workflow

Figure 2 shows the initial development comparison. SCI generating-point coverage was 0.965–0.995. The `ShapeChange` residual-bootstrap interval had zero coverage in 9 of 16 cells, all involving nonsmooth signals outside its documented smooth spline regime. In the four smooth logistic cells its coverage was 0.960–0.990 and its intervals were narrower. The correct use is therefore a hybrid: retain a strong S-shaped estimator as the point summary, and use SCI when the uncertainty claim must remain valid under nonsmoothness.

The hybrid did not improve point-estimation error. Aggregated mean absolute error was 0.06236 for the published S-shaped estimator and 0.06276 after a simple projection. We therefore make no "point-estimator booster" claim. The value of SCI is calibrated uncertainty around the point estimate.

8.4. Matched honest comparison

Table 3 compares SCI and the conservative PBP relaxation under the same Gaussian model and nominal coverage target. Across the 16 cells, the observed full-target coverage ranges were 0.950–0.995 for SCI and 0.995–1.000 for PBP; cellwise Wilson intervals are reported in the supplementary audit. For the cusp, onset, and jump, SCI reduced median width among each method's nonempty outputs by 19.4%–75.7%. For the weak logistic signal, both methods returned the same nearly full range. This experiment isolates the useful gain: direct feature-specific inversion can be much sharper than projecting a generic confidence box, but it cannot manufacture information when curvature is too weak.

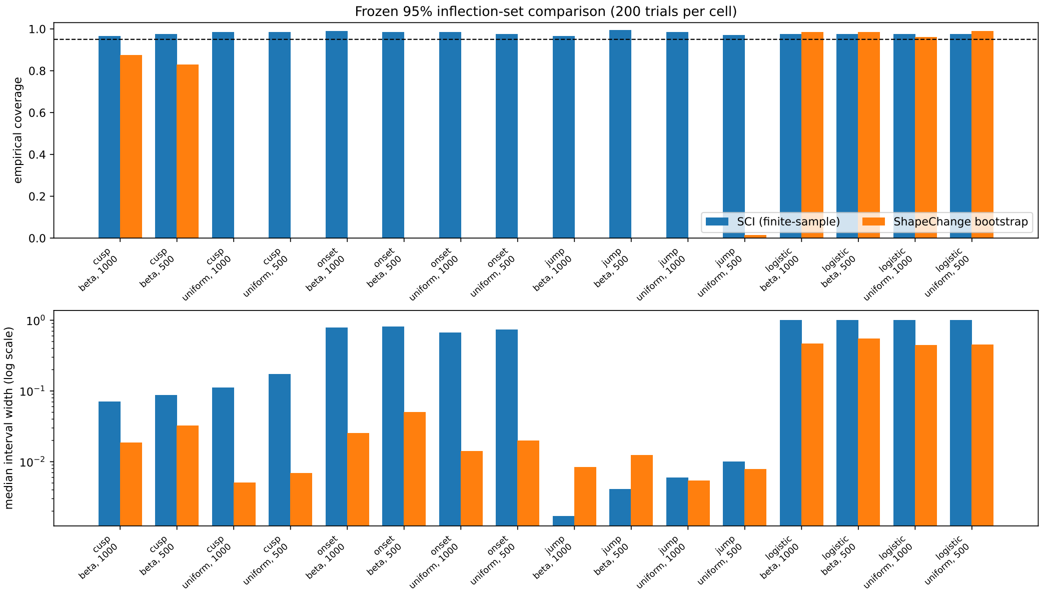

Figure 2. Coverage and width in the initial comparison. The smooth-bootstrap comparator is used inside and outside its smooth regime to expose the practical robustness gap; the comparison does not contradict its published assumptions.

Table 3. Matched honest comparison with 200 responses per cell. "Beta" is the Beta$(4, 8)$ design. Widths are medians among each method's nonempty outputs, and reduction compares these conditional medians. Wilson coverage intervals, empty rates, and a common-nonempty sensitivity analysis are reported in the supplementary audit.

Signal	Design	n	SCI cov.	PBP cov.	SCI width	PBP width	Reduction
Cusp	Uniform	500	.985	1.000	.1737	.4122	57.9%
Cusp	Uniform	1000	.965	1.000	.1109	.3861	71.3%
Cusp	Beta	500	.950	1.000	.0877	.3151	72.2%
Cusp	Beta	1000	.995	1.000	.0712	.2934	75.7%
Onset	Uniform	500	.975	.995	.7325	.9960	26.5%
Onset	Uniform	1000	.980	1.000	.6703	.9980	32.8%
Onset	Beta	500	.980	1.000	.5460	.6776	19.4%
Onset	Beta	1000	.985	1.000	.5170	.7116	27.3%
Jump	Uniform	500	.965	1.000	.0100	.0180	44.4%
Jump	Uniform	1000	.980	.995	.0050	.0090	44.4%
Jump	Beta	500	.975	.995	.0041	.0068	40.0%
Jump	Beta	1000	.985	1.000	.0017	.0034	50.0%
Logistic	Uniform	500	.970	1.000	.9960	.9960	0.0%
Logistic	Uniform	1000	.965	1.000	.9980	.9980	0.0%
Logistic	Beta	500	.975	1.000	.6776	.6776	0.0%
Logistic	Beta	1000	.975	1.000	.7116	.7116	0.0%

The post-audit paired bootstrap interval for width reduction has a positive lower endpoint in all 12 cusp, onset, and jump cells. Because this uncertainty summary was added after the frozen experiment, we treat it as descriptive rather than as a new decision gate.

SCI empty rates in this comparison range from 0 to 0.040, and PBP empty rates from 0 to 0.005. Restricting both methods to trials where both outputs are nonempty leaves the overall reduction range for the 12 informative cells at 19.4%–75.7%. The largest cellwise change is for the Beta-design jump at $n = 500$, where the reduction changes from 40.0% to 33.3%.

For this matched experiment, both procedures were restricted to the observed design range $[x_1, x_n]$. This matters for the random Beta design: its observed range is shorter than $[0, 1]$. The high-precision audit in Table 2 instead used the full scientific domain $[0, 1]$.

8.5. Unknown-scale and heteroskedastic checks

The unknown-homoskedastic experiment contained 28 frozen cells. The minimum SCI coverage was 0.990, the minimum coverage of the upper scale bound was 0.975, and the mean width inflation among informative cells was 1.117 relative to known scale. All frozen gates passed.

The bounded-heteroskedastic experiment contained 48 cells spanning constant, linear, and plume-like variance profiles. The minimum transition coverage and the minimum scale-envelope coverage were both 0.995. These results test the implementation of Theorems 6.1 and 6.3; the guarantees themselves come from the theorems, not from Monte Carlo performance.

9. Real-data illustrations

9.1. Atmospheric LIDAR

The LIDAR example uses the 221-observation `lidar` data set from the `SemiPar` package, originally reported in an atmospheric spectroscopy collection [15,16]. It studies a response curve over altitude. A published S-shaped point fit places the transition at 588 m. The `ShapeChange` fit gives 606 m with a residual-bootstrap interval $[592.6, 612.8]$ m. Visible variance changes make that iid-bootstrap interval descriptive here.

Figure 3 reports the SCI sensitivity analysis. With $\kappa = 1$, the set is $[510, 664]$ m. It grows to $[486, 720]$ m for $\kappa = 1.5$, $[438, 720]$ m for $\kappa = 2$, and $[390, 720]$ m for $\kappa \geq 3$ in the displayed range. The conclusion is conditional: if the maximum variance is at most κ times the mean variance, the corresponding set has the finite-sample guarantee. The widening shows the cost of making a more cautious variance claim.

9.2. Replicated DNase assay

The DNase ELISA data distributed with R contain 11 assay runs and eight concentrations; their documented sources are Davidian and Giltinan [14] and Pinheiro and Bates [17]. We use all 176 raw optical-density measurements [18]. Technical duplicates are averaged within each run, and the run is the independent unit. This leaves 11 replicate curves on the same eight-point design.

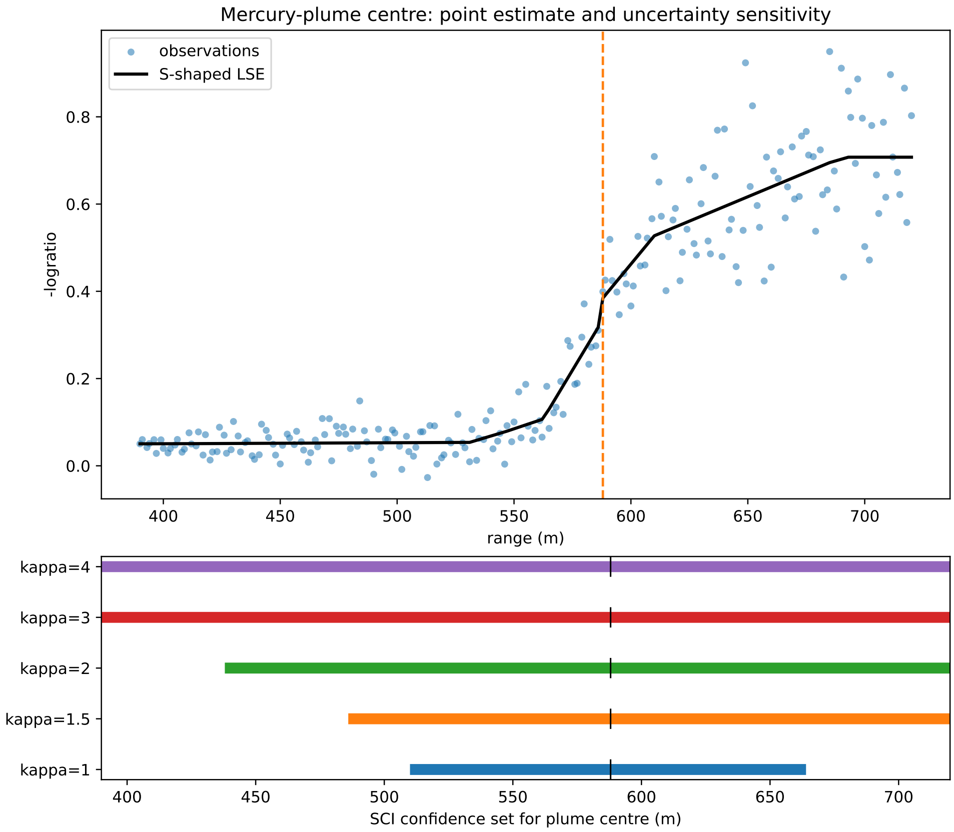

Figure 3. LIDAR transition sets under the declared variance-ratio bound κ. Larger heterogeneity allowances produce wider honest sets.

Figure 4 shows the replicate curves and the resulting uncertainty statement. The replicate Student construction gives the 95% transition set $[0.78125, 12.5]$ in concentration units. A four-parameter logistic fit gives the descriptive midpoint 4.1412. The point lies inside the confidence set, but the set reaches the largest observed concentration. The data support a lower limit, while they do not supply a reliable upper limit inside the observed range.

The exact statement assumes independent Gaussian replicate curves with a common mean and covariance. With only 11 runs, the interval is necessarily conservative. Treating all 176 readings as independent would give a narrower but invalid answer because technical duplicates and concentrations from the same run are correlated.

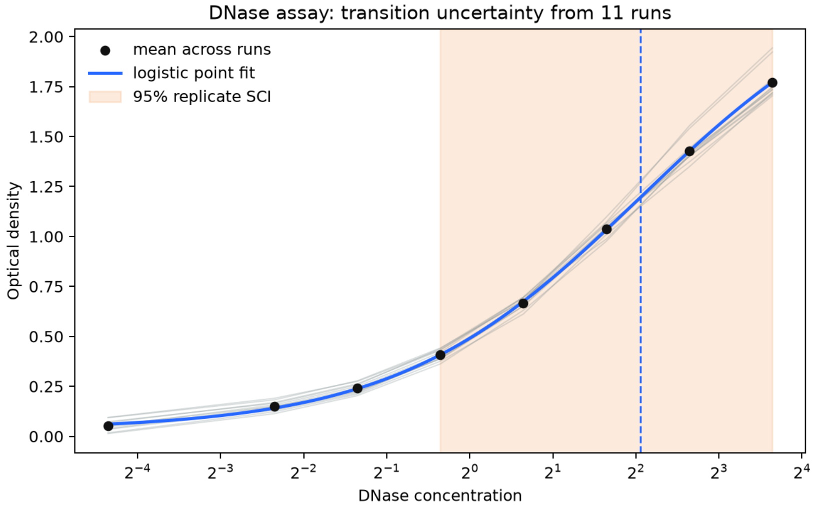

Figure 4. DNase run means, replicate mean curve, descriptive logistic midpoint, and the 95% replicate-curve SCI set. Dependence across concentrations inside a run is allowed.

10. Discussion

10.1. What practical problem is solved?

The method solves a specific uncertainty problem: it gives a finite-sample outer confidence set for every convex-to-concave transition admitted by at least one shape-constrained continuation of the sampled mean vector, without assuming a smooth or unique inflection. A researcher can attach this set to an S-shaped point estimate and distinguish three outcomes:

(1) a narrow set, meaning that the data certify local shape signs close to the fitted transition;
(2) a one-sided or wide set, meaning that only part of the location is identified; and
(3) an empty set, which may motivate a model check while remaining a controlled type-I event under the assumed model.

This addresses the gap exposed by the experiments. A smooth bootstrap can be sharp when its assumptions hold but fail badly on nonsmooth signals. A generic honest confidence region remains valid but can be unnecessarily wide. SCI keeps the broad non-smooth guarantee while using feature-specific evidence to improve width in informative cases.

10.2. What is not claimed?

First, SCI is not a new point estimator and did not improve point-estimation error in our hybrid experiment. Second, it is not the first inference method for inflection points. Third, the $(\log n/n)^{1/7}$ cubic rate is not new. Fourth, the output is an outer interval and can include points that no compatible continuation uses; it is not a proof that every included point is a true physical threshold. Fifth, PBP is a conservative split relaxation, not the sharpest projection of its pointwise band. Finally, the real-data conclusions are conditional on their sampling models.

10.3. Limitations and open work

The exact single-curve results are Gaussian. A self-normalized or distribution-robust band would broaden the method. The heteroskedastic extension needs a user-declared κ; it cannot learn an unrestricted variance vector from one observation per design point. The replicate result needs independent runs with a common covariance. Bonferroni calibration is simple and transparent but can be conservative when many contrasts overlap.

The main theoretical gap is also clear. Theorem 5.4 gives a rate under explicit contrast-margin and availability conditions. A broad function-class lemma that derives these conditions automatically, together with a matching lower bound for the set-valued target, remains open. Such a result could turn the current conditional rate into a minimax characterization. A stronger computational comparator would project a simultaneous band onto one globally joined convex-to-concave sequence rather than the split relaxation used here. A same-contrast-band version would also separate the effect of the inversion geometry from the choice of simultaneous band.

11. Conclusion

Shape-contrast inversion turns local geometric evidence into an honest uncertainty statement for a convex-to-concave transition. Its main strength is not a universally narrow interval. It is the combination of finite-sample coverage, tolerance of nonsmooth and nonunique transitions, and a direct matrix-free implementation. The matched comparison shows that targeting the feature can be much sharper than this generic confidence-box relaxation. The logistic and DNase examples show the equally important other side: when the data do not support precise localization, the method says so.

Data availability statement

The code and simulation results supporting this study are openly available at `https://github.com/Safreliy/hcrd`. The repository contains the matrix-free implementation, frozen experiment configurations, raw trial-level outputs, summaries, figures, and verification manifests under `results/sci` and `results/hct`. The DNase table and its source manifest are provided under `data/external/dnase`; the original data are distributed with R [18]. The LIDAR data are available in the `SemiPar` R package [16]; retrieval and analysis instructions are included with the experiment code. All reported numerical claims are generated from these artifacts. The test suite includes SCI-specific regression tests, and the artifact verifier checks the frozen SCI files against their manifests.

Commands for rebuilding this manuscript are given beside the source.

Funding

No funding was received for this work.

CRediT authorship contribution statement

Saveliy Baturin: Conceptualization; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Visualization; Writing – original draft; Writing – review and editing.

Disclosure of interest

The author reports there are no competing interests to declare.

Declaration of generative AI use

The author used OpenAI Codex to assist with literature organization, software engineering, verification, and language editing. The mathematical claims, experimental design, reported results, and final interpretations were checked and approved by the author, who assumes full responsibility for the manuscript.

Notes on contributor

Saveliy Baturin is an independent researcher working on statistical inference and shape-constrained methods.

References

[1] Feng OY, Chen Y, Han Q, et al. Nonparametric, tuning-free estimation of S-shaped functions. J R Stat Soc Ser B Stat Methodol. 2022;84(4):1324–1352. Doi: `https://doi.org/10.1111/rssb.12481`.

[2] Davies PL, Kovac A, Meise M. Nonparametric regression, confidence regions and regularization. Ann Stat. 2009;37(5B):2597–2625. Doi: `https://doi.org/10.1214/07-AOS575`.

[3] Schmidt-Hieber J, Munk A, Dümbgen L. Multiscale methods for shape constraints in deconvolution: Confidence statements for qualitative features. Ann Stat. 2013;41(3):1299–1328. Doi: `https://doi.org/10.1214/13-AOS1089`.

[4] Meyer MC. Inference using shape-restricted regression splines. Ann Appl Stat. 2008; 2(3):1013–1033. Doi: `https://doi.org/10.1214/08-AOAS167`.

[5] Liao X, Meyer MC. Change-point estimation using shape-restricted regression splines. J Stat Plan Inference. 2017;188:8–21. Doi: `https://doi.org/10.1016/j.jspi.2017.03.007`.

[6] Dümbgen L, Spokoiny VG. Multiscale testing of qualitative hypotheses. Ann Stat. 2001; 29(1):124–152. Doi: `https://doi.org/10.1214/aos/996986504`.

[7] Dümbgen L. Optimal confidence bands for shape-restricted curves. Bernoulli. 2003;9(3):423–449. Doi: https://doi.org/10.3150/bj/1065444812.
[8] Cai TT, Low MG, Xia Y. Adaptive confidence intervals for regression functions under shape constraints. Ann Stat. 2013;41(2):722–750. Doi: https://doi.org/10.1214/12-AOS1068.
[9] Robertson T, Wright FT, Dykstra RL. Order restricted statistical inference. New York: Wiley; 1988.
[10] Frick K, Munk A, Sieling H. Multiscale change point inference. J R Stat Soc Ser B Stat Methodol. 2014;76(3):495–580. Doi: https://doi.org/10.1111/rssb.12047.
[11] Chernozhukov V, Hong H, Tamer E. Estimation and confidence regions for parameter sets in econometric models. Econometrica. 2007;75(5):1243–1284. Doi: https://doi.org/10.1111/j.1468-0262.2007.00794.x.
[12] Anderson TW. The integral of a symmetric unimodal function over a symmetric convex set and some probability inequalities. Proc Am Math Soc. 1955;6(2):170–176. Doi: https://doi.org/10.2307/2032333.
[13] Laurent B, Massart P. Adaptive estimation of a quadratic functional by model selection. Ann Stat. 2000;28(5):1302–1338. Doi: https://doi.org/10.1214/aos/1015957395.
[14] Davidian M, Giltinan DM. Nonlinear models for repeated measurement data. London: Chapman and Hall; 1995.
[15] Sigrist MW, editor. Air monitoring by spectroscopic techniques. (Chemical Analysis; Vol. 127). New York: Wiley; 1994.
[16] Ruppert D, Wand MP, Carroll RJ. Semiparametric regression. Cambridge: Cambridge University Press; 2003. Doi: https://doi.org/10.1017/CBO9780511755453.
[17] Pinheiro JC, Bates DM. Mixed-effects models in S and S-PLUS. New York: Springer; 2000. Doi: https://doi.org/10.1007/b98882.
[18] R Core Team. R: A language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing; 2025. Available from: https://www.R-project.org/.

Appendix A. Contrast construction

The implementation uses a fixed dyadic family. For a block size q, a start index s, and separation multiplier $g \in \{1, 2, 4\}$, it averages the q elementary chord residuals formed by the index triples $(s+j, s+gq+j, s+2gq+j)$, $j = 0, \ldots, q-1$. On a uniform design this is

$$Z_{s,q,g} = \frac{1}{q} \sum_{j=0}^{q-1} \left(\frac{1}{2} Y_{s+j} - Y_{s+gq+j} + \frac{1}{2} Y_{s+2gq+j} \right).$$

For irregular designs, the two endpoint coefficients are the barycentric weights in (3). In both cases the contrast is a nonnegative average of elementary chord residuals. Its coefficients sum to zero, so affine functions have zero contrast; convex functions give a nonnegative mean and concave functions a nonpositive mean.

At each (q, g), starts are spaced by hq, where the default stride multiplier is $h = 1$; the last admissible start is appended if it is not already on that grid. The family is fixed from the design before the responses are read. On a uniform design, prefix sums of Y_i evaluate each three-block average in constant time. The known-scale standard error follows from the closed-form squared norm of the row. Irregular designs use compact index and weight calculations rather than a dense $M \times n$ matrix.

Appendix B. Proofs

Proof of Lemma 3.3. Because $m_2 \in \mathcal{I}(g)$, the function g is convex on $[A, m_2]$, hence on $[m_1, m_2]$. Because $m_1 \in \mathcal{I}(g)$, it is concave on $[m_1, B]$, hence on the same interval. A real-valued function that is both convex and concave on an interval is affine there. For any $m \in [m_1, m_2]$, membership of m_2 gives convexity on $[A, m]$, while membership of m_1 gives concavity on $[m, B]$. Thus $m \in \mathcal{I}(g)$. This proves the interval property. Closedness does not follow. For example, on $[0, 1]$, let $g(x) = 0$ for $x < 1$ and $g(1) = -1$. Then $\mathcal{I}(g) = [0, 1)$: the downward endpoint jump is compatible with concavity, but it prevents convexity on the full interval. $\square$

Proof of Proposition 3.4. For a convex function, successive secant slopes are nondecreasing; for a concave function, they are nonincreasing. Thus the stated prefix and suffix conditions are necessary. If $m \in (x_k, x_{k+1})$, convexity on the left also requires $v \geq \mu_k + s_{k-1}(m - x_k)$ when $k > 1$, while concavity on the right requires $v \leq \mu_{k+1} - s_{k+1}(x_{k+1} - m)$ when $k < n - 1$. This proves necessity of (2).

Conversely, under these conditions, linearly interpolate the sampled points on each side and choose any v between the displayed bounds. The resulting left slopes are nondecreasing and the right slopes are nonincreasing, so the two piecewise-linear parts give a valid continuation meeting at m. Appropriate linear endpoint extensions cover $[A, x_1]$ and $[x_n, B]$ when needed. At a design point the same construction uses $v = \mu_i$. Prefix and suffix feasibility can each be scanned once. In a gap, (2) is one linear inequality in m, so its feasible portion and the extreme target endpoints require constant additional work.

For completeness, let p be the last design index with a feasible convex prefix and q the first index with a feasible concave suffix. These prefix and suffix index sets are respectively initial and terminal segments. A design point x_i is feasible exactly when $q \leq i \leq p$, while a gap can be feasible only when $q - 1 \leq i \leq p$. Every gap between two feasible design points is entirely feasible because its linear inequality holds at both endpoints. When $q \leq p$, each possible boundary-gap portion meets the nearest feasible point. When $q = p + 1$, only the gap $i = p$ can remain: if it is an interior gap, its linear inequality fails at both endpoints and hence throughout; if it is the first or last gap with one missing inequality, the whole gap is feasible. For $q > p + 1$ no gap is possible. Hence the union has no separated components. Taking the closure gives one closed interval, or the empty set, and completes the proof. $\square$

Proof of Theorem 5.1. For each T, $Z_T - \mu_T$ is centered normal with standard deviation $\sigma \left\| w_T \right\|_2$. By (4) and a union bound,

$$\mathrm{P}\{\ell_T \leq \mu_T \leq u_T \text{ for all } T \in \mathcal{T}\} \geq 1 - \alpha.$$

Call this event E and first fix an m in the unclosed set inside (1). Choose a compatible $g \in \mathcal{G}_x(\mu)$ with $m \in \mathcal{I}(g)$. If $\ell_T > 0$ and $a_T \geq m$, the support lies in the concave part of g, so $\mu_T \leq 0$. This contradicts $\ell_T \leq \mu_T$ on E. Therefore every certified positive contrast has $a_T < m$, and $\widehat{L} \leq m$. Likewise, if $u_T < 0$ and $b_T \leq m$, the support lies in the convex part, so $\mu_T \geq 0$, contradicting $\mu_T \leq u_T$. Hence $m \leq \widehat{U}$. The argument holds for every such m on the same event E. Since $[\widehat{L}, \widehat{U}]$ is closed, it also contains the closure $\mathcal{I}_x(\mu)$. Moreover, if the target is nonempty, the strict inequalities $a_T < m < b_T$ for active signs imply $\widehat{L} < \widehat{U}$ on E (with the inactive boundary cases following from $A < B$). Hence the algorithm returns this interval, not the empty set. The uniform bound follows because the probability of E does not depend on μ. $\square$

Proof of Proposition 5.2. A certified positive contrast rules out points at or below its left endpoint; a certified negative contrast rules out points at or above its right endpoint. Intersecting the resulting strict half-lines with $[A, B]$ gives the raw survivor set. If $\widehat{L} < \widehat{U}$, its closure is exactly $[\widehat{L}, \widehat{U}]$. If $\widehat{L} \geq \widehat{U}$, the two strict requirements are incompatible and the survivor set is empty. $\square$

Proof of Theorem 5.4. For the selected left contrast,

$$\mathrm{P}\left\{Z_{T_L} > \mu_{T_L} - z_{1-\eta/2}\sigma \left\|w_{T_L}\right\|_2\right\} \geq 1 - \eta/2,$$

and the analogous upper-tail statement holds for T_R. With probability at least $1 - \eta$, both selected rows are therefore certified: $\ell_{T_L} > 0$ and $u_{T_R} < 0$. On this event, $\widehat{L} \geq a_{T_L} \geq m_- - Kd$ and $\widehat{U} \leq b_{T_R} \leq m_+ + Kd$, which proves localization and the length bound without using simultaneous coverage.

The simultaneous band event from Theorem 5.1 has probability at least $1 - \alpha$ and gives $\mathcal{I}_x(\mu) \subseteq \widehat{\mathcal{C}}(Y)$. Intersecting it with the two certification events and applying a union bound gives the joint probability $1 - \alpha - \eta$. $\square$

Proof of Proposition 7.1. With probability at least $1 - \alpha$, the Bonferroni box contains the complete sampled mean vector μ. Fix a transition in the unclosed set inside (1) and choose a compatible continuation g. The sampled values on the left form a feasible convex prefix, and those on the right form a feasible concave suffix. Thus the PBP split relaxation retains that transition. This includes transitions outside the observed range because its two boundary cuts use the declared endpoints A and B. The returned set is closed, so it also contains the closure $\mathcal{I}_x(\mu)$. Bonferroni proves the claim. $\square$

Proof of Corollary 5.5. For a symmetric equally spaced triple with middle location u and physical separation d, direct expansion of the cubic gives

$$Q(f) = 3B(m_0 - u)d^2. \tag{B1}$$

Take an implemented multiplier-one row of block size q. Its q triples have physical separation $d = q/(n + 1)$ and use disjoint left, middle, and right blocks. Its coefficient norm is therefore exactly

$$\|w_T\|_2 = \sqrt{\frac{3}{2q}}. \tag{B2}$$

Choose the rightmost allowed start, on the q-spaced grid of starts, whose whole support ends at or before m_0. The interior condition ensures that it exists. Every triple has its right point at or before m_0, so its middle point is at least d to the left of m_0. Equation (B1) gives $\mu_T \geq 3Bd^3$. Maximality of the start also puts its support beginning less than $4d$ to the left of m_0. With (B2),

$$\frac{\mu_T}{\sigma \left\|w_T\right\|_2} \geq \frac{\sqrt{6}B\sqrt{n+1}\, d^{7/2}}{\sigma} \geq c_{\alpha, M_n} + z_{1-\eta/2},$$

where the last step uses $q \geq q_*$. This row fails to be certified positive with probability at most $\eta/2$.

The symmetric construction uses the leftmost q-grid start whose support begins at or after m_0. It gives a negative row with failure probability at most $\eta/2$ and support ending less than $4d$ to the right of m_0. When both rows are certified, $\widehat{L} > m_0 - 4d$ and $\widehat{U} < m_0 + 4d$. The smallest-dyadic choice gives $q < 2q_*$, so $d < 2h_*$ and $\widehat{U} - \widehat{L} < 16h_*$. The two-row certification event alone has probability at least $1 - \eta$, proving the first assertion. Intersecting it with the simultaneous-band event of Theorem 5.1 retains the whole set $\mathcal{I}_x(\mu)$ and gives total probability at least $1 - \alpha - \eta$.

At each dyadic q and separation multiplier, there are at most $n/q + 2$ starts. Summing n/q over dyadic sizes gives less than $2n$. With three multipliers and at most $1 + \log_2 n$ sizes, $M_n \le 6n + 6(1 + \log_2 n)$. Thus $c_{\alpha,M_n} = O(\sqrt{\log n})$, which gives the stated order. □

Proof of Theorem 6.1. Write $R_X Y = R_X\mu + \sigma R_X G$ with $G \sim N(0, I)$. The set $\{v : \|v\|_2^2 \le q\}$ is convex and centrally symmetric in the residual subspace. Anderson's inequality implies that translating a centered Gaussian cannot increase its probability of lying in this set. Hence

$$\underset{\mu,\sigma}{\mathrm{P}}\left\{\frac{\|R_X Y\|_2^2}{\sigma^2} \le q_{n-r,\eta}\right\} \le \mathrm{P}\{\chi^2_{n-r} \le q_{n-r,\eta}\} = \eta.$$

Thus $\overline{\sigma} \ge \sigma$ with probability at least $1 - \eta$. On this event, bands using $\overline{\sigma}$ are no narrower than valid bands using σ. Combining this event with a contrast-band error budget $\alpha - \eta$ proves the result by a union bound. □

Proof of Theorem 6.3. Let $D = \mathrm{diag}(\sigma_1^2, \dots, \sigma_n^2)$ and $A_D = D^{1/2} R D^{1/2}$. For $G \sim N(0, I_n)$, the centered quadratic form is

$$Q_0 = \left\|RD^{1/2}G\right\|_2^2 = G^\top A_D G = \sum_j \lambda_j G_j^2,$$

where the λ_j are the nonnegative eigenvalues of A_D. Anderson's inequality for the centered Gaussian vector $RD^{1/2}G$ and the Euclidean ball in the range of R gives

$$\mathrm{P}\{Q \le t\} \le \mathrm{P}\{Q_0 \le t\}, \qquad t \ge 0. \tag{B3}$$

If observation i belongs to a block of size $b_i \in \{2, 3\}$, then $R_{ii} = 1 - 1/b_i \ge 1/2$. Therefore

$$\mathrm{tr}(A_D) = \mathrm{tr}(RD) = \sum_i R_{ii}\sigma_i^2 \ge \frac{n\bar{\sigma}^2}{2}. \tag{B4}$$

Because R is an orthogonal projector and Assumption 6.2 holds,

$$\lambda_{\max}(A_D) \le \max_i \sigma_i^2 \le \kappa\bar{\sigma}^2, \qquad \mathrm{tr}(A_D^2) \le \lambda_{\max}(A_D)\,\mathrm{tr}(A_D).$$

Together with (B4), these inequalities give $\lambda_{\max}(A_D)/\,\mathrm{tr}(A_D) \le 2\kappa/n$. The lower-tail inequality of Laurent and Massart [13], with $x = \log(1/\eta)$, now yields with probability at least $1 - \eta$,

$$\begin{aligned} Q_0 &\geq \operatorname{tr}(A_D) - 2\sqrt{x \operatorname{tr}(A_D^2)} \\ &\geq \operatorname{tr}(A_D)\left(1 - 2\sqrt{\frac{2\kappa x}{n}}\right) \\ &\geq \frac{n\bar{\sigma}^2}{2} d_{n,\kappa,\eta}. \end{aligned}$$

The last step uses $d_{n,\kappa,\eta} > 0$. Combining this bound with (B3) gives the same lower bound for Q. Hence

$$\widehat{\sigma}^2_{\max} = \frac{2\kappa Q}{n d_{n,\kappa,\eta}} \geq \kappa\bar{\sigma}^2 \geq \max_i \sigma_i^2$$

with probability at least $1 - \eta$. On this event every contrast standard error is bounded above by $\widehat{\sigma}_{\max} \|w_T\|_2$. Combining it with the contrast-band event of probability at least $1 - (\alpha - \eta)$ proves the coverage statement by a union bound. ☐

Proof of Theorem 6.4. For each fixed j, the replicate contrasts $Z_{1j}, \ldots, Z_{N_{\text{rep}},j}$ are iid univariate Gaussian. The usual Student statistic therefore has a $t_{N_{\text{rep}}-1}$ distribution whenever its variance is positive. In a zero-variance direction the sample mean equals the contrast mean almost surely, so the degenerate interval is exact. A Bonferroni union bound over the M two-sided intervals gives simultaneous coverage at least $1 - \alpha$. The deterministic sign-inversion argument from Theorem 5.1 completes the proof. ☐

Appendix C. Additional experimental detail

All stochastic experiments use recorded seeds and frozen JSON configurations. The high-precision audit was run independently of the development simulations. For each cell, the deterministic endpoints of $\mathcal{I}_x(\mu)$ were computed by Proposition 3.4. Coverage is the fraction of saved SCI sets whose left endpoint is no larger than $\inf \mathcal{I}_x(\mu)$ and whose right endpoint is no smaller than $\sup \mathcal{I}_x(\mu)$. This postprocessing did not regenerate any response. Empty sets count as noncoverage and are reported separately; width summaries use the nonempty sets. The E38 sensitivity audit also reports medians on the common subset where both procedures are nonempty.

Table C1 gives the deterministic targets used in the corrected E36 coverage calculation. The E38 observed-range domains give the same internal target endpoints for these cells.

Table C2 reports the E38 empty rates and the requested same-trial sensitivity analysis. "Conditional" uses each method's own nonempty outputs, as in Table 3. "Shared" uses only trials where both outputs are nonempty.

The PBP computation uses two families of linear feasibility problems. One finds the longest prefix on which some vector in the pointwise box has nondecreasing divided slopes. The other finds the earliest suffix on which some, potentially different, vector has nonincreasing divided slopes. Monotonicity of feasibility permits binary search. The resulting outer range is a conservative discrete split relaxation because the two witnesses are not required to join at a common transition value.

Table C1. Design-identified transition sets $\mathcal{I}_x(\mu)$ for the 16 benchmark cells in E36. Here n is the number of observations and Beta denotes the Beta$(4, 8)$ design. Endpoints are in the signal's input coordinates.

Signal	Design	n	$\mathcal{I}_x(\mu)$
Cusp	Uniform	500	$[0.299103, 0.301696]$
Cusp	Uniform	1000	$[0.299551, 0.300849]$
Cusp	Beta	500	$[0.299491, 0.300419]$
Cusp	Beta	1000	$[0.299673, 0.300225]$
Onset	Uniform	500	$[0.300000, 0.304618]$
Onset	Uniform	1000	$[0.300000, 0.302311]$
Onset	Beta	500	$[0.300000, 0.301350]$
Onset	Beta	1000	$[0.300000, 0.300608]$
Jump	Uniform	500	$[0.299401, 0.301397]$
Jump	Uniform	1000	$[0.299700, 0.300699]$
Jump	Beta	500	$[0.299558, 0.300239]$
Jump	Beta	1000	$[0.299705, 0.300045]$
Logistic	Uniform	500	$[0.297779, 0.302811]$
Logistic	Uniform	1000	$[0.298889, 0.301407]$
Logistic	Beta	500	$[0.299071, 0.300778]$
Logistic	Beta	1000	$[0.299472, 0.300350]$

Table C2. E38 empty-output rates (proportions over 200 responses per cell) and percentage reductions in median width for SCI relative to PBP. "Conditional" uses each method's own nonempty outputs; "Shared" uses the same trials where both methods are nonempty. Beta denotes the Beta$(4, 8)$ design, and n is the number of observations.

Signal	Design	n	SCI empty	PBP empty	Conditional	Shared
Cusp	Uniform	500	.010	.000	57.9%	58.1%
Cusp	Uniform	1000	.030	.000	71.3%	71.5%
Cusp	Beta	500	.040	.000	72.2%	72.1%
Cusp	Beta	1000	.005	.000	75.7%	75.7%
Onset	Uniform	500	.005	.000	26.5%	26.5%
Onset	Uniform	1000	.000	.000	32.8%	32.8%
Onset	Beta	500	.005	.000	19.4%	19.4%
Onset	Beta	1000	.000	.000	27.3%	27.3%
Jump	Uniform	500	.035	.000	44.4%	44.4%
Jump	Uniform	1000	.020	.005	44.4%	44.4%
Jump	Beta	500	.025	.005	40.0%	33.3%
Jump	Beta	1000	.015	.000	50.0%	50.0%
Logistic	Uniform	500	.000	.000	0.0%	0.0%
Logistic	Uniform	1000	.000	.000	0.0%	0.0%
Logistic	Beta	500	.000	.000	0.0%	0.0%
Logistic	Beta	1000	.000	.000	0.0%	0.0%